

PROJECT PROPOSAL

COURSE: Mobile Application Development
COURSE ID: 119528

PROJECT TITLE: Interactive Quiz App with AI Suggestions

GROUP MEMBERS

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1 Overview

This project is a mobile application that allows users to attempt quizzes on different topics. It also uses AI to provide hints, explanations, and suggestions to improve learning.

2 Introduction

Mobile applications are widely used for learning and practice. Quiz apps help users test their knowledge, but most apps do not provide proper explanations or guidance. This project aims to build a smart quiz app that not only tests users but also helps them learn using AI.

3 Problem Statement

Many quiz applications only show correct or incorrect answers without explaining them. Users do not understand their mistakes, which reduces learning effectiveness. There is a need for a system that provides proper feedback and guidance to users.

4 Goals/Objectives

The main objectives of this project are:

- To develop a mobile quiz application
- To provide multiple quiz categories
- To track user scores and performance
- To integrate AI for explanations

5 Solution

The proposed solution is an interactive quiz app built using React Native. The app will allow users to attempt quizzes and receive AI-based suggestions. Firebase will be used for authentication and database, and a free AI API (Groq API) will be used to generate explanations.

6 Technology Stack

The following technologies will be used to develop this project:

- **Frontend:** React Native for building the mobile application
- **Backend:** Firebase for authentication
- **AI Integration:** Groq API for generating explanations
- **API Integration:** Trivia API for fetching quiz questions
- **Navigation:** React Navigation for managing different screens

7 Procedures/Scope of Work

The project will be completed in the following steps:

- Design user interface of the app
- Implement quiz functionality
- Integrate Firebase for login and data storage
- Add AI features such as explanations
- Test the application and fix errors

8 Evaluation

The project will be evaluated based on:

- Functionality of the app
- User interface and experience
- Accuracy of quiz system
- Performance of AI suggestions

9 Conclusion

This project will provide an interactive and intelligent learning platform. It will help users improve their knowledge through quizzes and AI-based feedback. It also demonstrates modern mobile app development techniques.